

Class-2

1. Finite Fourier cosine transformation of $f(t)$ in the interval $(0, \pi)$, c_n is _____

- a) $\int_{-\pi}^{\pi} f(t) \cos(nt) dt$ b) $2 \int_0^{\pi} f(t) \cos(nt) dt$
c) $\int_0^{\pi} f(t) \sin(nt) dt$ d) $\int_0^{\pi} f(t) \cos(nt) dt$

Ans: d

2. Finite Fourier sine transformation of $f(t)$ in the interval $(0, l)$, s_n is _____

- a) $\int_0^l f(t) \sin\left(\frac{n\pi t}{l}\right) dt$ b) $\int_0^l f(t) \cos\left(\frac{n\pi t}{l}\right) dt$
c) $\int_{-l}^l f(t) \sin\left(\frac{n\pi t}{l}\right) dt$ d) $2 \int_0^l f(t) \sin\left(\frac{n\pi t}{l}\right) dt$

Ans: a

3. Finite Fourier cosine transformation of $f(t)=1$ in the interval $(0, \pi)$, c_n is _____

- a) π b) zero
c) $\cos(n\pi)$ d) $\sin(n\pi)$

Ans: b

4. Finite Fourier sine transform of $f(t)=1$ in, $S_n(0, \pi)$ is _____

- a) $\frac{1+(-1)^n}{n}$ b) $\frac{(-1)^n - 1}{n}$
c) $\frac{1-(-1)^n}{n}$ d) $1 - \frac{(-1)^n}{n}$

Ans: c

5. If the Fourier sine transform of $f(t)$, $S_n = \frac{1 - \cos n\pi}{n^2 \pi^2}$, $(0 \leq x \leq \pi)$ then the inverse

Fourier sine transform $f(t)$ is _____

- a) $\frac{2}{\pi} \sum_{n=1}^{\infty} \left(\frac{1 - \cos n\pi}{n^2} \right) \sin(nt)$ b) $\frac{2}{\pi^3} \sum_{n=1}^{\infty} \left(\frac{1 - \cos n\pi}{n^2} \right) \sin(nt)$
c) $\frac{\pi}{2} \sum_{n=1}^{\infty} \left(\frac{1 - \cos n\pi}{n^2} \right) \sin(nt)$ d) $\frac{\pi}{2} \sum_{n=1}^{\infty} \left(\frac{1 - \cos n\pi}{n^2} \right) \cos(nt)$

Ans: b

6. Inverse Fourier cosine transform of $f(t)$ in the interval $(0, l)$ is _____

- a) $\frac{1}{l} C_0 + \frac{2}{l} \sum_{n=1}^{\infty} C_n \cos(nt)$ b) $\frac{2}{l} \sum_{n=1}^{\infty} C_n \cos(nt)$
c) $\frac{1}{l} C_0 - \frac{2}{l} \sum_{n=1}^{\infty} C_n \cos(nt)$ d) $\frac{1}{l} C_0 + 2 \sum_{n=1}^{\infty} C_n \cos(nt)$

Ans: a